

INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des Sciences Technologies

TD No 4 – Reseaux 1

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Soumis au professeur : Ismael SAINT AMOUR

DATE: 20 novembre 2025

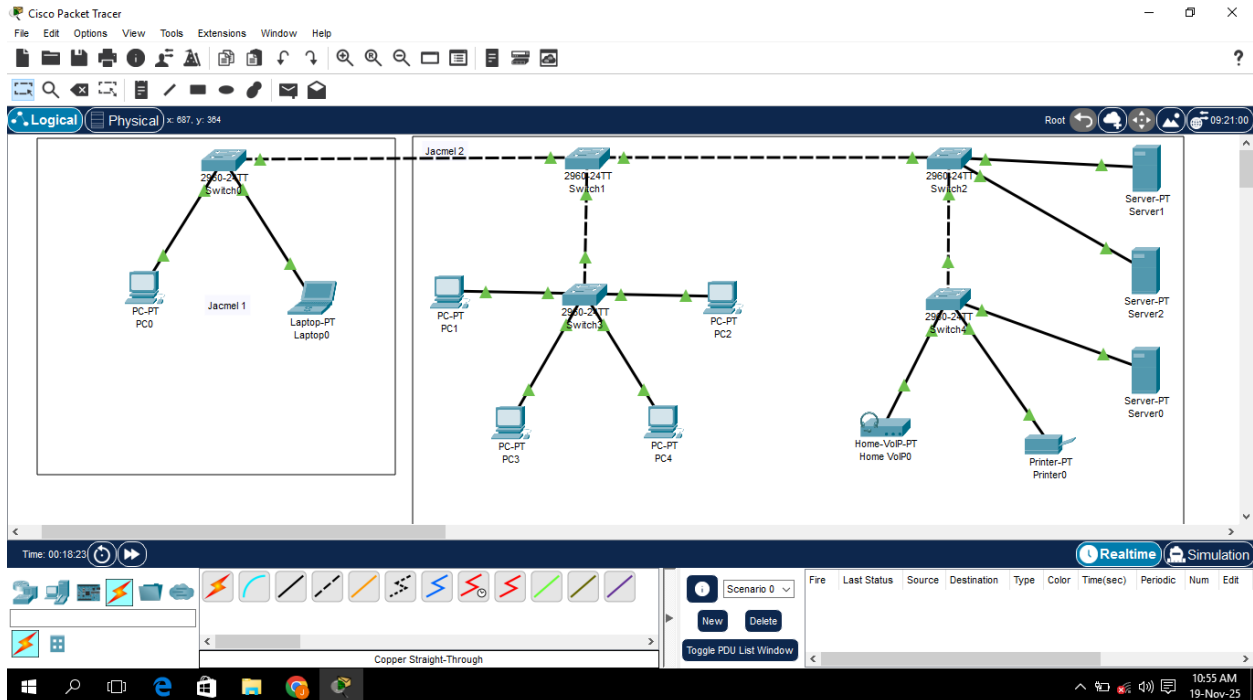
TD 4

Configuration d'un switch et d'un routeur avec Cisco Packet Tracer & Mode Simulation

L'objectif de ce TD est de:

- 1. Découvrir la configuration de base d'un switch et d'un routeur.**
- 2. Attribuer des adresses IPv4 et IPv6 aux périphériques.**
- 3. Tester la connectivité entre les hôtes.**
- 4. Utiliser le Mode Simulation pour analyser le trafic réseau.**

1) Reproduction de la topologie 1



2) Configuration des switches IPv4

Pour configurer les switches on fait un clique sur le switch en question et on va dans "CLI", une fois la nouvelle fenetre s'ouvre on commence la configuration avec les commandes suivantes

```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW1
Switch(config)# interface vlan 1
Switch(config-if) # ip address 192.168.1.1 255.255.255.0
Switch(config-if) # no shutdown
Switch(config)# exit
Switch(config)# end
```

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 431, y: 0

2950-24T Switch0

PC-PT PC0 Jacmel 1 Laptop-PT Laptop0

Time: 00:44:06

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
base1(config)#interface Vlan1
base1(config-if)#ip address 192.168.1.1 255.255.255.0
base1(config-if)#no shutdown

base1(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
base1(config-if)#exit
base1(config)#end
base1#
%SYS-5-CONFIG_I: Configured from console by console

base1#enable
base1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
base1(config)#interface vlan2
base1(config-if)#ip address 2001:db8:1::1 /64

% Invalid input detected at '^' marker.

base1(config-if)#ip address 2001:db8:1::fd/64

% Invalid input detected at '^' marker.

base1(config-if)# ip address 2001:db8:1::fc/64

% Invalid input detected at '^' marker.

base1(config-if)#ip address 2001:db8:1::fc/64
```

Copy Paste

Root

Server-PT Server1

Server-PT Server2

Server-PT Server0

Printer-PT Printer0

Realtime Simulation

Type Color Time(sec) Periodic Num Edit

11:22 AM 19-Nov-25

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 688, y: 19

2950-24T Switch0

PC-PT PC0 Jacmel 1 Laptop-PT Laptop0

Time: 00:44:21

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

```
Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname base2
base2(config)#interface Vlan1
base2(config-if)#ip address 192.168.2.1 255.255.255.0
base2(config-if)#no shutdown

base2(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
base2(config-if)#exit
base2(config)#end
base2#
%SYS-5-CONFIG_I: Configured from console by console
```

Copy Paste

Root

Server-PT Server1

Server-PT Server2

Server-PT Server0

Printer-PT Printer0

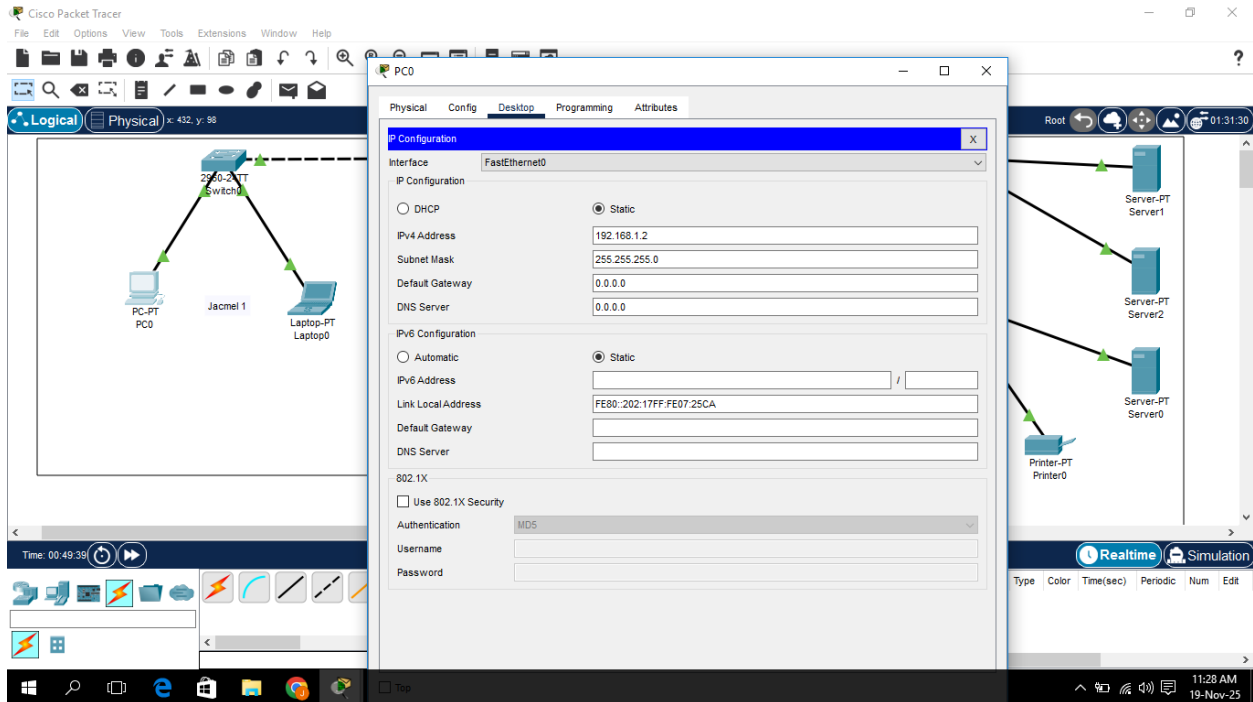
Realtime Simulation

Type Color Time(sec) Periodic Num Edit

11:22 AM 19-Nov-25

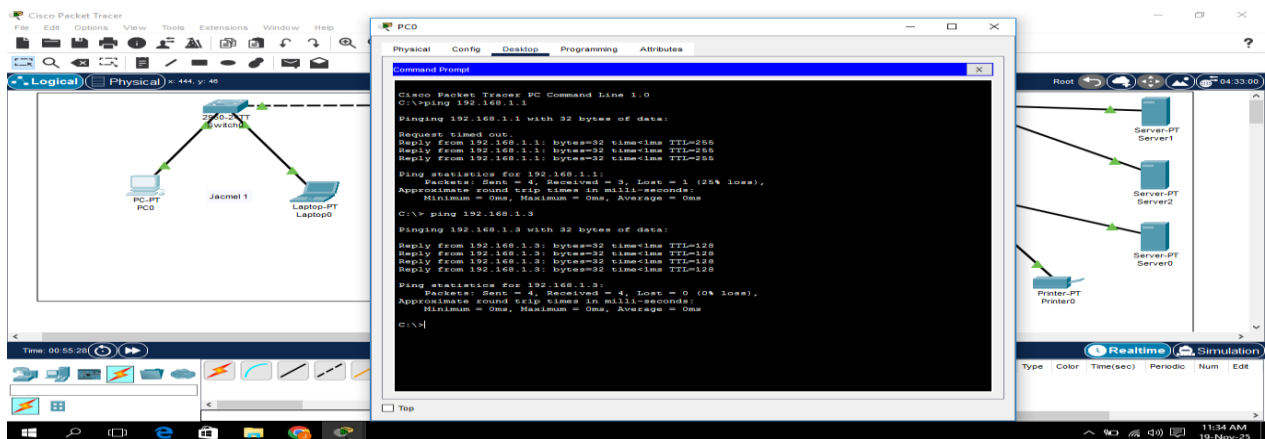
Configuration des dispositifs

Pour configurer les les dispositifs comme les Pc, laptop et autres, on fait un clique sur l'appareil on va dans Desktop apres dans ip configure et on met l'adresse IP, ainsi on fait de meme pour tout les autres dispositifs

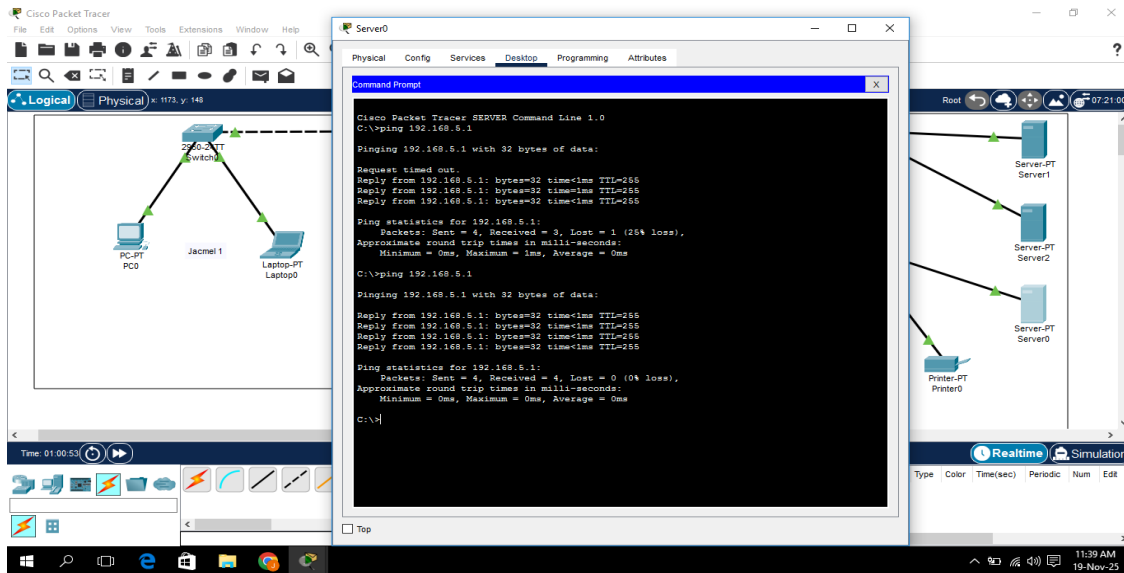
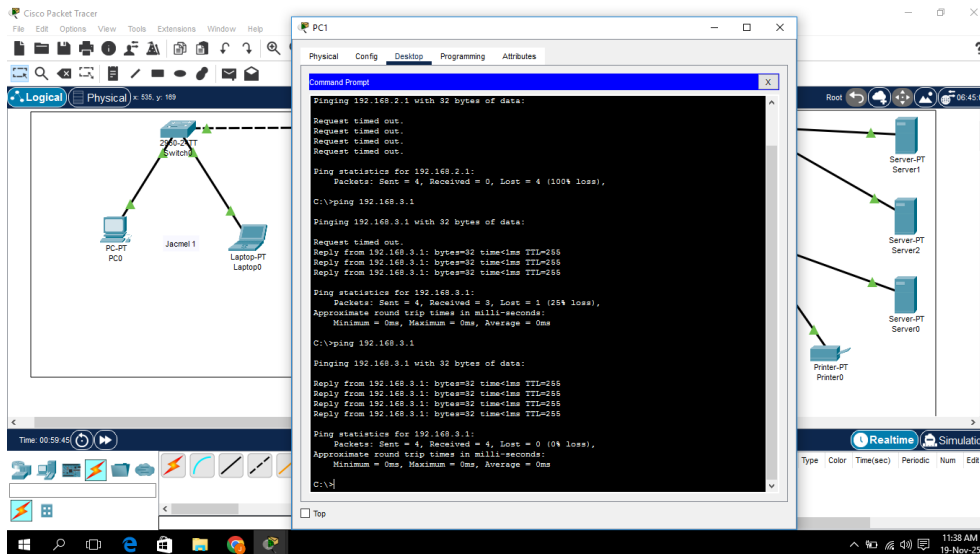


Tester la connectivité entre les hôtes

Pour tester la connectivité entre les Hotes on fait un clique sur l'appareil on va dans Desktop apres dans command Prompt , pour faire le test on ping plus l'adresse IP de l'appareil en question

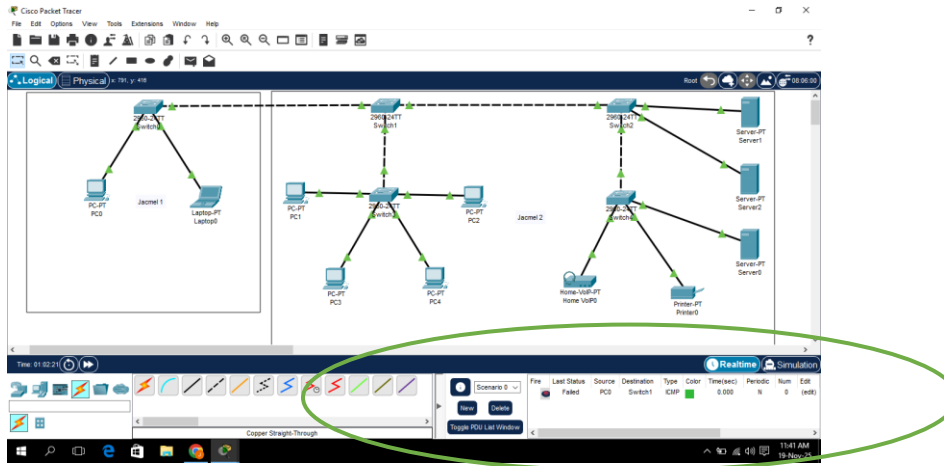


Ici on peut voir qu'il y a pas de connection car il y a pas de passerelle entre les reseaux

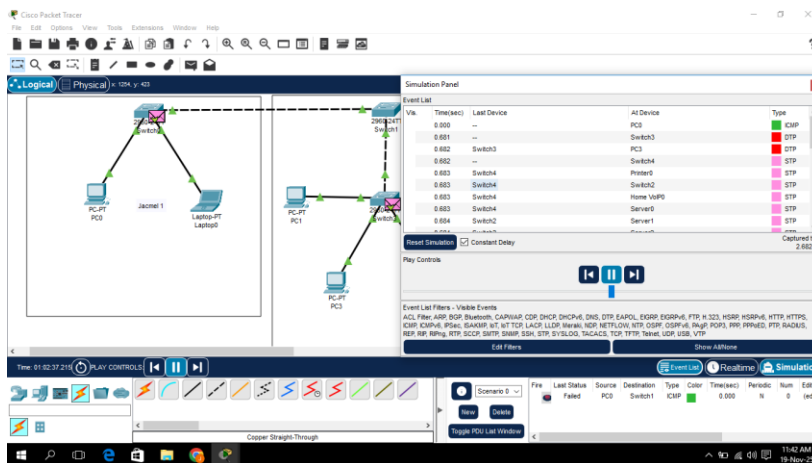


Mode de simulation

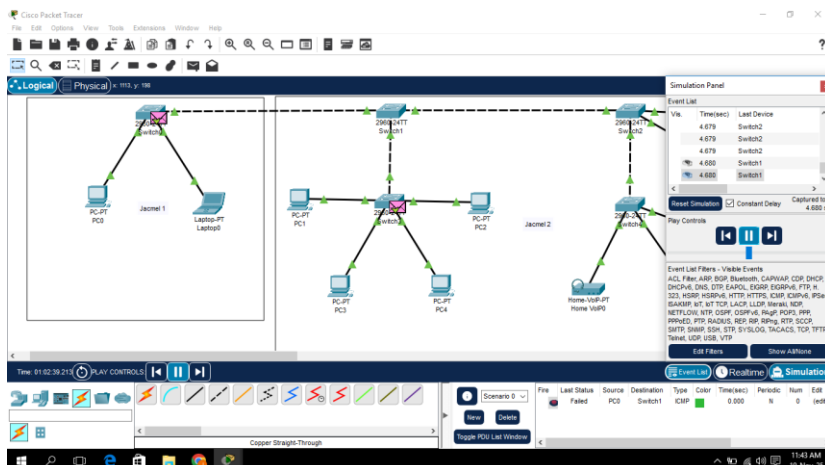
Pour tester les reseaux dans la mode de simulation on un message dans l'un des deux reseaux e suivons la demarche

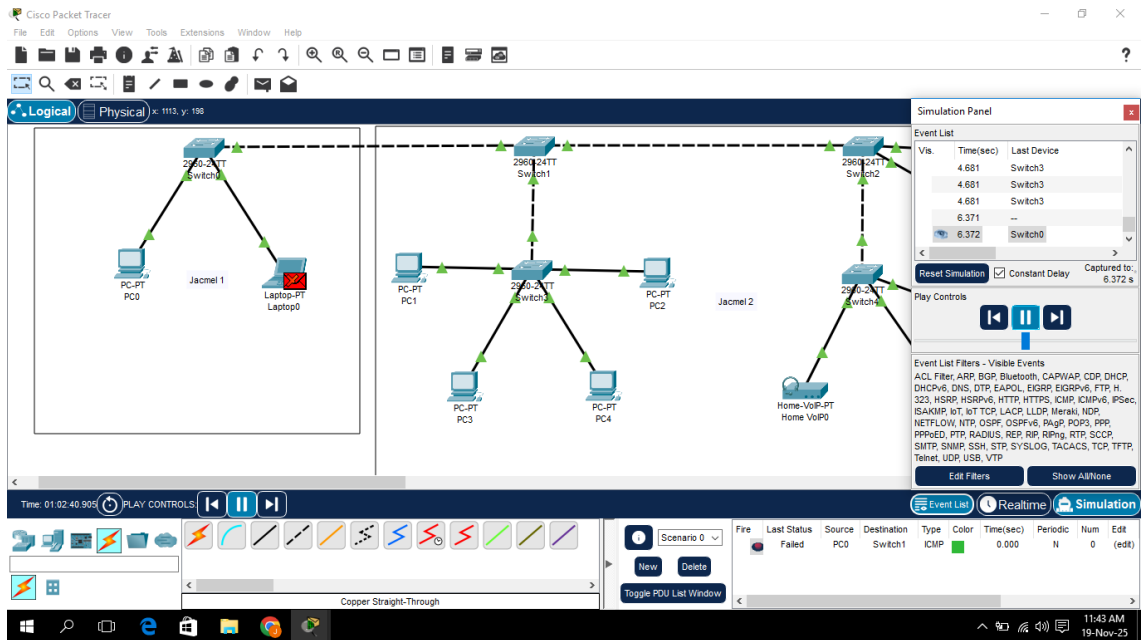
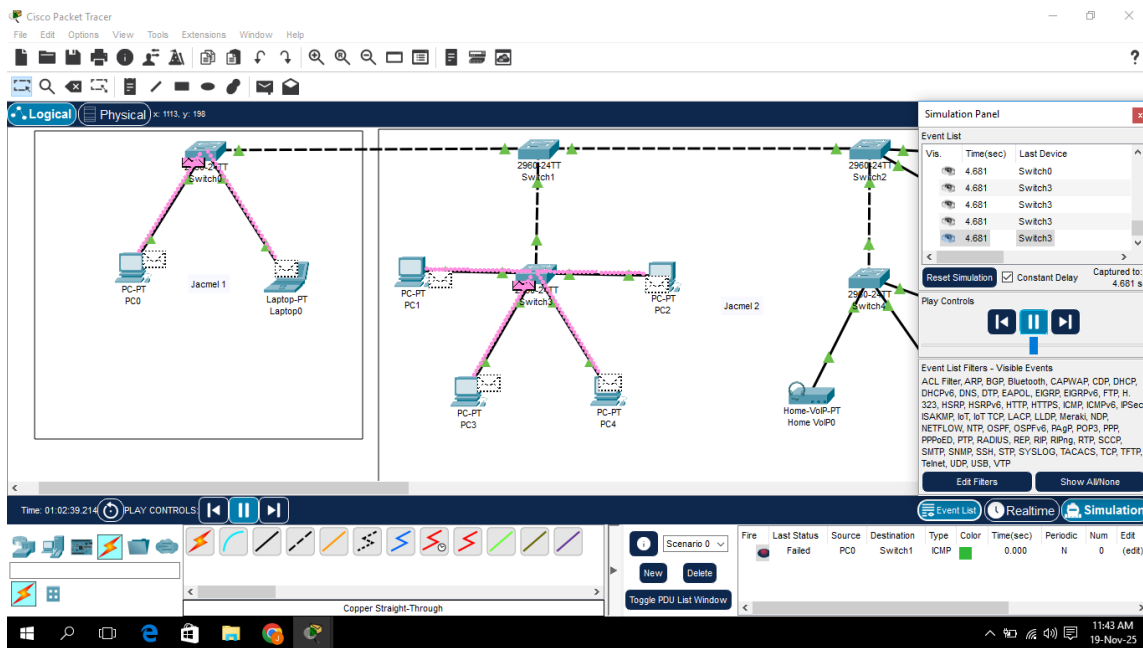


On ouvre le controle panel pour voir comment le paquet circule



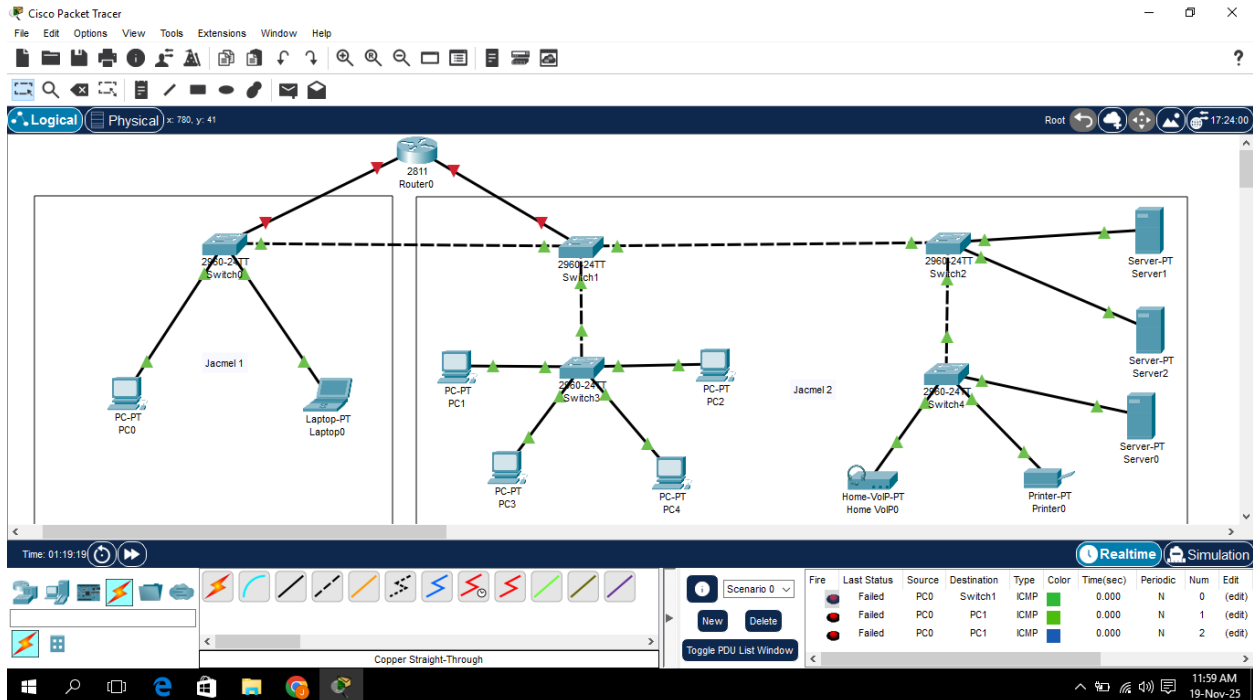
On peut aussi suivre le deplacement des paquets dans le schema





En Conclusion pour cette topologie les paquets ne circulent pas tres bien car il y a pas de router et de passerelle pour connecter les deux reseaux entre eux.

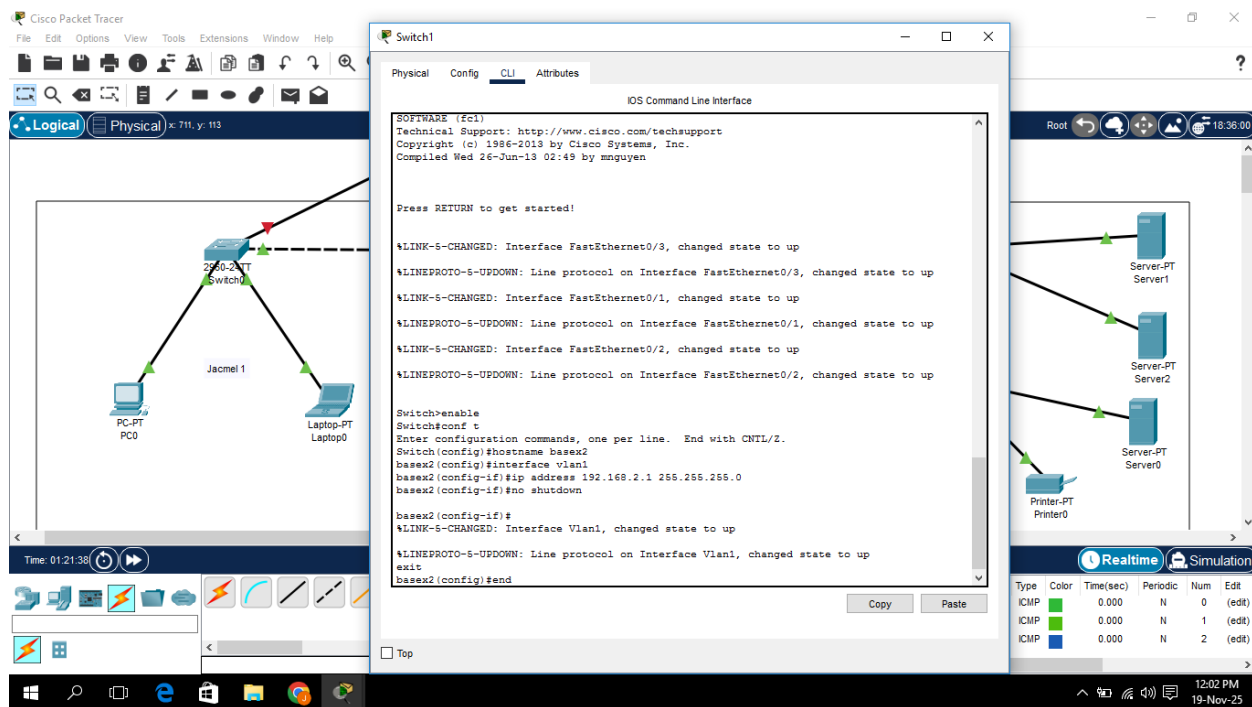
2) Reproduction de la topologie 2



Configuration des Switchs

Pour configurer les switchs on fait un clique sur le switch en question et on va dans "CLI", une fois la nouvelle fenetre s'ouvre on commence la configuration avec les commandes suivantes

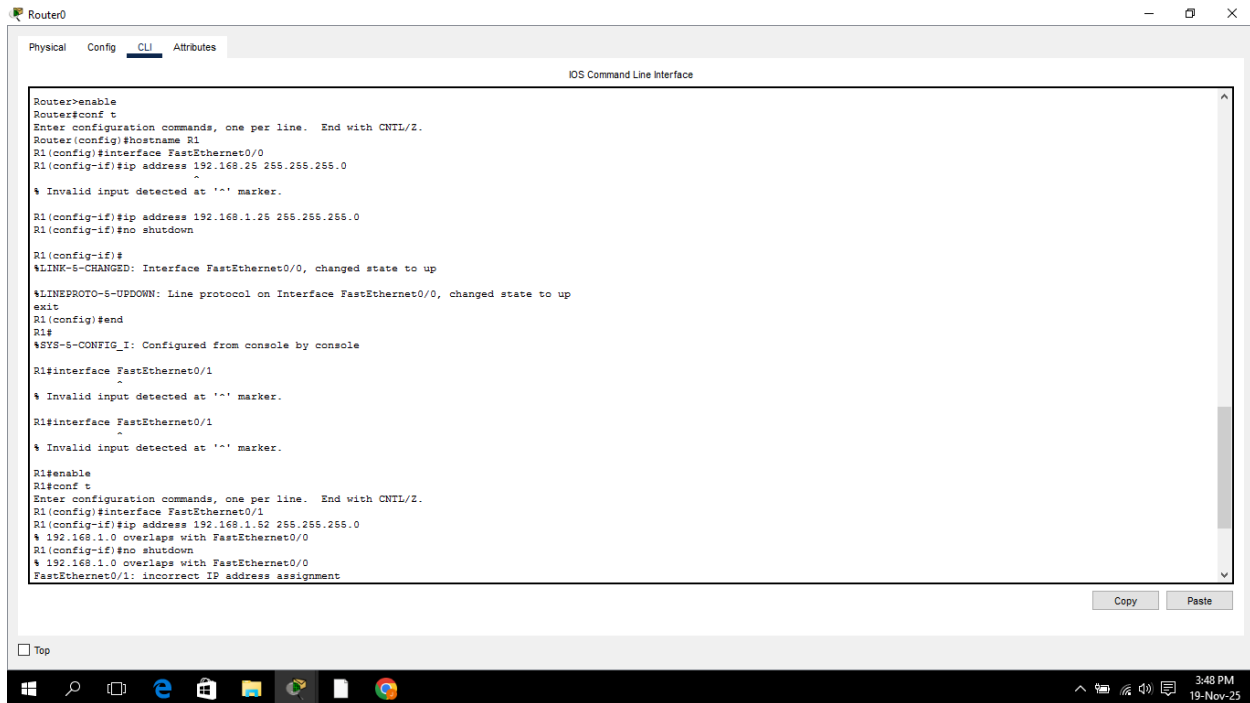
```
Switch> enable
Switch# configure terminal
Switch(config)# hostname SW1
Switch(config)# interface vlan 1
Switch(config-if) # ip address 192.168.1.1 255.255.255.0
Switch(config-if) # no shutdown
Switch(config)# exit
Switch(config)# end
```



Configuration du routeur

Pour configurer les switches on fait un clique sur le switch en question et on va dans "CLI", une fois la nouvelle fenetre s'ouvre on commence la configuration avec les commandes suivantes

```
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface FastEthernet0/0
R1(config-if) # ip address 192.168.1.254 255.255.255.0
R1(config-if) # no shutdown
R1(config-if) # exit
R1(config)# end
R1(config)# interface FastEthernet0/1
R1(config-if) # ip address 192.168.1.354 255.255.255.0
R1(config-if) # no shutdown
R1(config-if) # exit
R1(config)# end
```



```
Router>enable
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.1.25 255.255.255.0
^
% Invalid input detected at '^' marker.

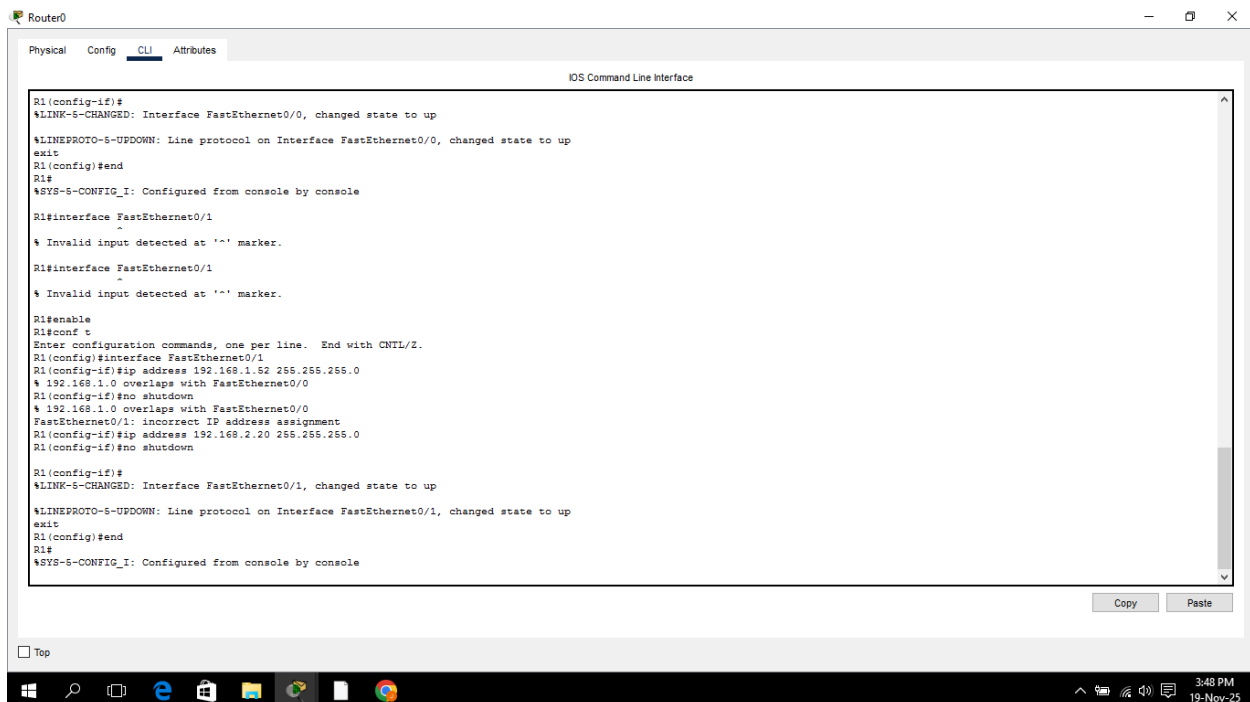
R1(config-if)#ip address 192.168.1.25 255.255.255.0
R1(config-if)#no shutdown
^
% Invalid input detected at '^' marker.

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#interface FastEthernet0/1
^
% Invalid input detected at '^' marker.

R1#interface FastEthernet0/1
^
% Invalid input detected at '^' marker.

R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet0/1
R1(config-if)#ip address 192.168.1.52 255.255.255.0
^
% 192.168.1.0 overlaps with FastEthernet0/0
R1(config-if)#no shutdown
^
% 192.168.1.0 overlaps with FastEthernet0/0
FastEthernet0/1: incorrect IP address assignment
R1(config-if)#no shutdown
```



```
R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#interface FastEthernet0/1
^
% Invalid input detected at '^' marker.

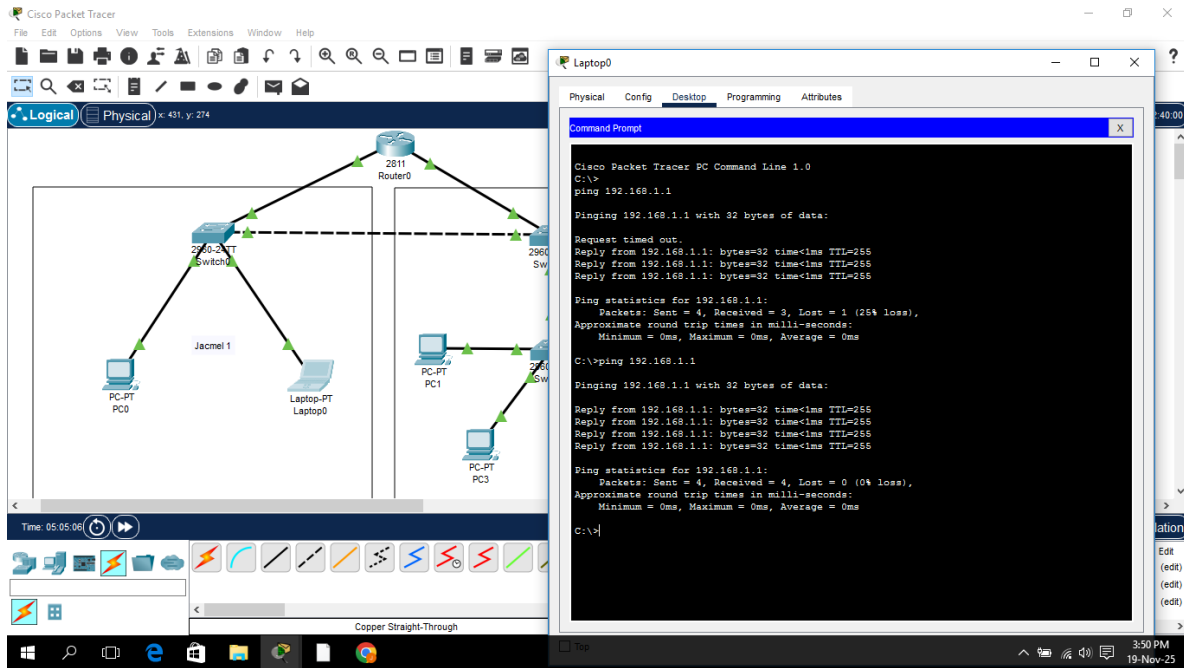
R1#interface FastEthernet0/1
^
% Invalid input detected at '^' marker.

R1#enable
R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet0/1
R1(config-if)#ip address 192.168.1.52 255.255.255.0
^
% 192.168.1.0 overlaps with FastEthernet0/0
R1(config-if)#no shutdown
^
% 192.168.1.0 overlaps with FastEthernet0/0
FastEthernet0/1: incorrect IP address assignment
R1(config-if)#ip address 192.168.2.20 255.255.255.0
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console
```

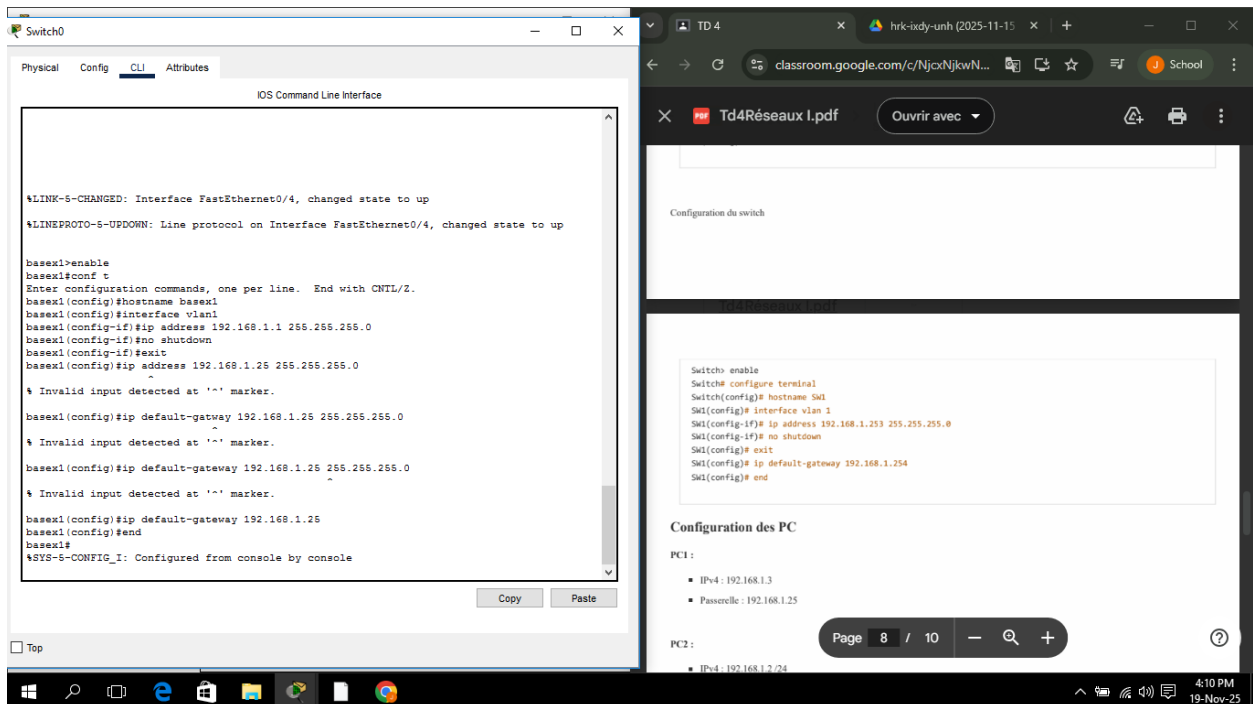
Testons la connectivité

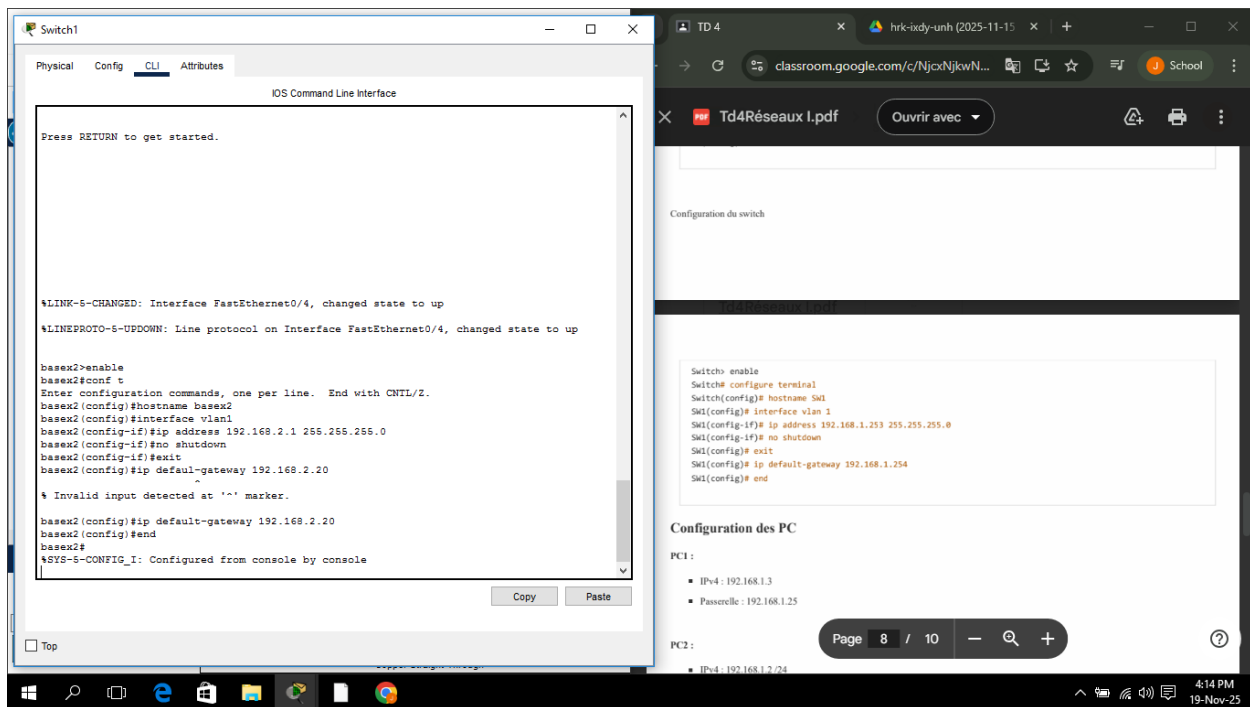
Pour tester la connectivité entre les Hotes on fait un clique sur l'appareil on va dans Desktop apres dans command Prompt, pour faire le test on ping plus l'adresse IP de l'appareil en question



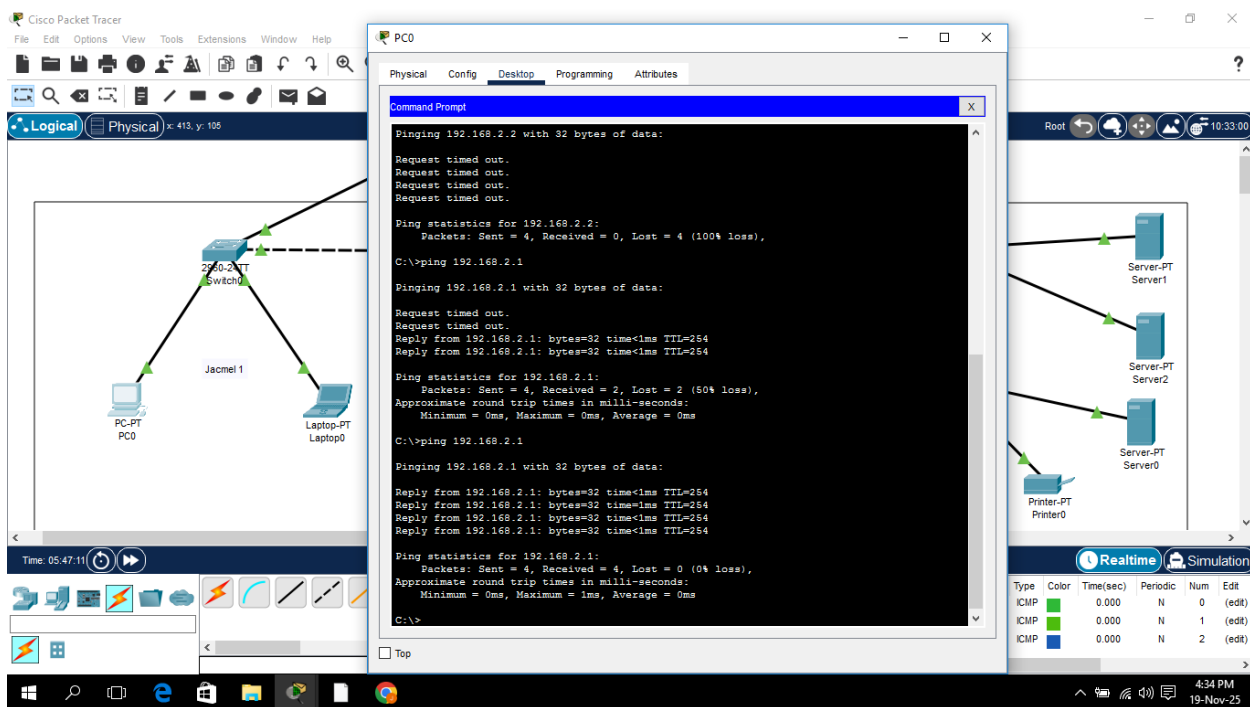
Mettons les passerelles

Pour mettre les passerelles on fait les codes dans le terminale du switch en question





Testons de nouveau la connectivité



Maintenant testons avec mode de simulation

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1142, y: 243

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.233	Switch3	PC3	STP
	0.233	Switch3	PC1	STP
	0.233	Switch3	PC4	STP
	0.233	Switch3	PC2	STP
	2.228	Switch4	Switch4	STP
	2.229	Switch4	Printer0	STP
	2.229	Switch4	Switch2	STP
	2.229	Switch4	Home VoIP0	STP
	2.229	Switch4	Server0	STP

Reset Simulation ☒ Constant Delay Capturing...

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PaGP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telet, UDP, USB, VTP

Edit Filters Show All/None

Time: 05:47:46.479 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

4:36 PM 19-Nov-25

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 1142, y: 243

Simulation Panel

Event List

Vis.	Time(sec)	Last Device	At Device	Type
	0.233	Switch3	PC2	STP
	2.228	Switch4	Switch4	STP
	2.229	Switch4	Printer0	STP
	2.229	Switch4	Switch2	STP
	2.229	Switch4	Home VoIP0	STP
	2.229	Switch4	Server0	STP
	2.230	Switch2	Server1	STP
	2.230	Switch2	Server2	STP
	2.230	Switch2	Switch1	STP

Reset Simulation ☒ Constant Delay Capturing...

Play Controls

Event List Filters - Visible Events

ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PaGP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telet, UDP, USB, VTP

Edit Filters Show All/None

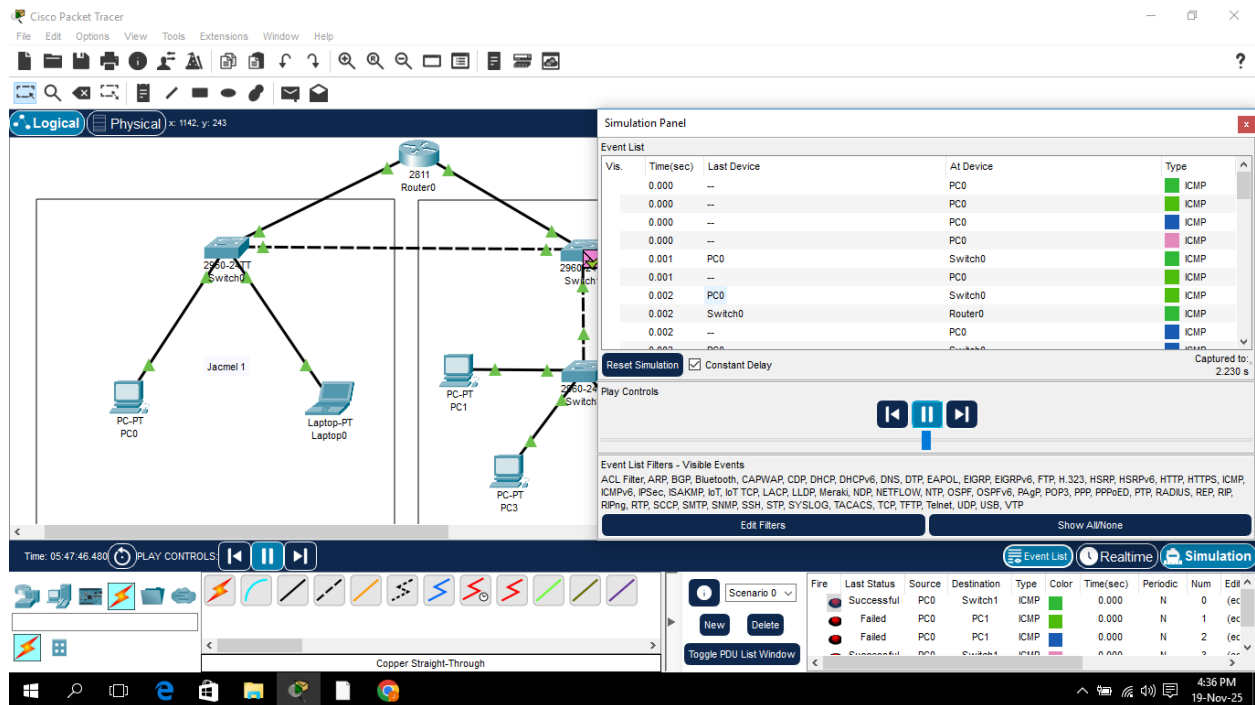
Time: 05:47:46.480 PLAY CONTROLS

Scenario 0

New Delete

Toggle PDU List Window

4:36 PM 19-Nov-25



En conclusion les pasquets sont bien distribuer car il y a des passerelle entre les reseaux donc les reseaux sont connectés entre eux.